

314707003_10.18.R

CDH

2026-04-29

```
rm(list=ls())  
#install.packages("AER")  
library(P0E5Rdata)
```

```
##  
## 載入套件：'P0E5Rdata'
```

```
## 下列物件被遮斷自 'package:datasets':  
##  
##      euro
```

```
library(AER)
```

```
## Warning: 套件 'AER' 是用 R 版本 4.5.3 來建造的
```

```
## 載入需要的套件：car
```

```
## Warning: 套件 'car' 是用 R 版本 4.5.3 來建造的
```

```
## 載入需要的套件：carData
```

```
## Warning: 套件 'carData' 是用 R 版本 4.5.3 來建造的
```

```
## 載入需要的套件：lmtest
```

```
## Warning: 套件 'lmtest' 是用 R 版本 4.5.3 來建造的
```

```
## 載入需要的套件：zoo
```

```
## Warning: 套件 'zoo' 是用 R 版本 4.5.3 來建造的
```

```
##  
## 載入套件：'zoo'
```

```
## 下列物件被遮斷自 'package:base':  
##  
##      as.Date, as.Date.numeric
```

```
## 載入需要的套件 : sandwich
```

```
## Warning: 套件 'sandwich' 是用 R 版本 4.5.3 來建造的
```

```
## 載入需要的套件 : survival
```

```
library(car)  
data = mroz  
  
data("mroz")  
  
dat <- subset(mroz, lfp == 1)  
  
# (a)  
dat$mothercoll <- ifelse(dat$mothereduc > 12, 1, 0)  
dat$fathercoll <- ifelse(dat$fathereduc > 12, 1, 0)  
  
cat("Sample size:", nrow(dat), "\n")
```

```
## Sample size: 428
```

```
cat("Mother some college percentage:", mean(dat$mothercoll) * 100, "\n")
```

```
## Mother some college percentage: 12.14953
```

```
cat("Father some college percentage:", mean(dat$fathercoll) * 100, "\n")
```

```
## Father some college percentage: 11.68224
```

```
table(dat$mothercoll)
```

```
##  
##      0      1  
## 376    52
```

```
table(dat$fathercoll)
```

```
##  
##      0      1
```

```
## 378 50
```

```
# (b)
cor(dat[, c("educ", "mothercoll", "fathercoll")])
```

```
##                educ mothercoll fathercoll
## educ          1.0000000  0.3594705  0.3984962
## mothercoll    0.3594705  1.0000000  0.3545709
## fathercoll    0.3984962  0.3545709  1.0000000
```

```
# optional comparison
cor(dat[, c("educ", "mothereduc", "fathereduc")])
```

```
##                educ mothereduc fathereduc
## educ          1.0000000  0.3870198  0.4154030
## mothereduc    0.3870198  1.0000000  0.5540632
## fathereduc    0.4154030  0.5540632  1.0000000
```

```
# (c)
iv_c <- ivreg(
  log(wage) ~ educ + exper + I(exper^2) |
  exper + I(exper^2) + mothercoll,
  data = dat
)

summary(iv_c)
```

```
##
## Call:
## ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | exper +
##       I(exper^2) + mothercoll, data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.08719 -0.32444  0.04147  0.36634  2.35621
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.1327561  0.4965325  -0.267  0.78932
## educ          0.0760180  0.0394077   1.929  0.05440 .
## exper         0.0433444  0.0134135   3.231  0.00133 **
## I(exper^2)    -0.0008711  0.0004017  -2.169  0.03066 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6703 on 424 degrees of freedom
## Multiple R-Squared: 0.147, Adjusted R-squared: 0.1409
## Wald test: 8.2 on 3 and 424 DF, p-value: 2.569e-05
```

```
confint(iv_c, "educ", level = 0.95)
```

```
##           2.5 %    97.5 %
## educ -0.001219763 0.1532557
```

```
# (d)
fs_d <- lm(
  educ ~ mothercoll + exper + I(exper^2),
  data = dat
)

summary(fs_d)
```

```
##
## Call:
## lm(formula = educ ~ mothercoll + exper + I(exper^2), data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.4267 -0.4826 -0.3731  1.0000  4.9353
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 12.079094   0.303118  39.849 < 2e-16 ***
## mothercoll   2.517068   0.315713   7.973 1.46e-14 ***
## exper        0.056230   0.042101   1.336  0.182
## I(exper^2)  -0.001956   0.001256  -1.557  0.120
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.133 on 424 degrees of freedom
## Multiple R-squared:  0.1347, Adjusted R-squared:  0.1285
## F-statistic: 21.99 on 3 and 424 DF,  p-value: 2.965e-13
```

```
linearHypothesis(fs_d, "mothercoll = 0")
```

```
##
## Linear hypothesis test:
## mothercoll = 0
##
## Model 1: restricted model
## Model 2: educ ~ mothercoll + exper + I(exper^2)
##
##      Res.Df    RSS Df Sum of Sq      F      Pr(>F)
## 1         425 2219.2
## 2         424 1929.9   1     289.32 63.563 1.455e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
t_mothercoll <- summary(fs_d)$coefficients["mothercoll", "t value"]
cat("First-stage F in part (d):", t_mothercoll^2, "\n")
```

```
## First-stage F in part (d): 63.56307
```

```
# (e)
iv_e <- ivreg(
  log(wage) ~ educ + exper + I(exper^2) |
  mothercoll + fathercoll + exper + I(exper^2),
  data = dat
)

summary(iv_e)
```

```
##
## Call:
## ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | mothercoll +
##       fathercoll + exper + I(exper^2), data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.07797 -0.32128  0.03418  0.37648  2.36183
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.2790819   0.3922213  -0.712   0.47714
## educ         0.0878477   0.0307808   2.854   0.00453 **
## exper        0.0426761   0.0132950   3.210   0.00143 **
## I(exper^2)   -0.0008486   0.0003976  -2.135   0.03337 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6679 on 424 degrees of freedom
## Multiple R-Squared: 0.153, Adjusted R-squared: 0.147
## Wald test: 9.724 on 3 and 424 DF, p-value: 3.224e-06
```

```
confint(iv_e, "educ", level = 0.95)
```

```
##           2.5 %    97.5 %
## educ 0.02751845 0.1481769
```

```
# (f)
fs_f <- lm(
  educ ~ mothercoll + fathercoll + exper + I(exper^2),
  data = dat
)

summary(fs_f)
```

```
##
## Call:
## lm(formula = educ ~ mothercoll + fathercoll + exper + I(exper^2),
##     data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.2152 -0.3056 -0.2152  0.7627  5.0620
##
```

```
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 11.890259   0.290251  40.965 < 2e-16 ***
## mothercoll   1.749947   0.322347   5.429 9.58e-08 ***
## fathercoll   2.186612   0.329917   6.628 1.04e-10 ***
## exper        0.049149   0.040133   1.225  0.221
## I(exper^2)   -0.001449   0.001199  -1.209  0.227
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.033 on 423 degrees of freedom
## Multiple R-squared:  0.2161, Adjusted R-squared:  0.2086
## F-statistic: 29.15 on 4 and 423 DF,  p-value: < 2.2e-16
```

```
linearHypothesis(
  fs_f,
  c("mothercoll = 0", "fathercoll = 0")
)
```

```
##
## Linear hypothesis test:
## mothercoll = 0
## fathercoll = 0
##
## Model 1: restricted model
## Model 2: educ ~ mothercoll + fathercoll + exper + I(exper^2)
##
##      Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1      425 2219.2
## 2      423 1748.3  2    470.88 56.963 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# (g)
summary(iv_e, diagnostics = TRUE)
```

```
##
## Call:
## ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | mothercoll +
##       fathercoll + exper + I(exper^2), data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.07797 -0.32128  0.03418  0.37648  2.36183
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.2790819   0.3922213  -0.712  0.47714
## educ         0.0878477   0.0307808   2.854  0.00453 **
## exper        0.0426761   0.0132950   3.210  0.00143 **
## I(exper^2)   -0.0008486   0.0003976  -2.135  0.03337 *
##
## Diagnostic tests:
##              df1 df2 statistic p-value
## Weak instruments  2 423    56.963 <2e-16 ***
```

```
## Wu-Hausman          1 423      0.519    0.472
## Sargan              1 NA       0.238    0.626
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6679 on 424 degrees of freedom
## Multiple R-Squared: 0.153,    Adjusted R-squared: 0.147
## Wald test: 9.724 on 3 and 424 DF,  p-value: 3.224e-06
```


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```
rm(list=ls())  
#install.packages("AER")  
library(P0E5Rdata)
```

```
##  
## 載入套件：'P0E5Rdata'
```

```
## 下列物件被遮斷自 'package:datasets':  
##  
##      euro
```

```
library(AER)
```

```
## Warning: 套件 'AER' 是用 R 版本 4.5.3 來建造的
```

```
## 載入需要的套件：car
```

```
## Warning: 套件 'car' 是用 R 版本 4.5.3 來建造的
```

```
## 載入需要的套件：carData
```

```
## Warning: 套件 'carData' 是用 R 版本 4.5.3 來建造的
```

```
## 載入需要的套件：lmtest
```

```
## Warning: 套件 'lmtest' 是用 R 版本 4.5.3 來建造的
```

```
## 載入需要的套件：zoo
```

```
## Warning: 套件 'zoo' 是用 R 版本 4.5.3 來建造的
```

```
##  
## 載入套件：'zoo'
```

```
## 下列物件被遮斷自 'package:base':  
##  
##      as.Date, as.Date.numeric
```

```
## 載入需要的套件: sandwich
```

```
## Warning: 套件 'sandwich' 是用 R 版本 4.5.3 來建造的
```

```
## 載入需要的套件: survival
```

```
library(car)  
  
data("capm5")  
dat = capm5  
  
dat$MSFT_RF <- dat$msft - dat$riskfree  
dat$MKT_RF  <- dat$mkt  - dat$riskfree  
  
# (a) OLS CAPM  
ols_a <- lm(MSFT_RF ~ MKT_RF, data = dat)  
summary(ols_a)
```

```
##  
## Call:  
## lm(formula = MSFT_RF ~ MKT_RF, data = dat)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -0.27424 -0.04744 -0.00820  0.03869  0.35801   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)  0.003250   0.006036   0.538   0.591      
## MKT_RF       1.201840   0.122152   9.839  <2e-16 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 0.08083 on 178 degrees of freedom  
## Multiple R-squared:  0.3523, Adjusted R-squared:  0.3486   
## F-statistic: 96.8 on 1 and 178 DF,  p-value: < 2.2e-16
```

```
confint(ols_a)
```

```
##              2.5 %      97.5 %  
## (Intercept) -0.008661409 0.01516061  
## MKT_RF       0.960788169 1.44289134
```

```
beta_ols <- coef(ols_a)["MKT_RF"]  
cat("OLS beta =", beta_ols, "\n")
```

```
## OLS beta = 1.20184
```

```
# (b) RANK and first stage
dat$RANK <- rank(dat$MKT_RF, ties.method = "first")

fs_b <- lm(MKT_RF ~ RANK, data = dat)
summary(fs_b)
```

```
##
## Call:
## lm(formula = MKT_RF ~ RANK, data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.110497 -0.006308  0.001497  0.009433  0.029513
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -7.903e-02  2.195e-03  -36.0   <2e-16 ***
## RANK         9.067e-04  2.104e-05   43.1   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.01467 on 178 degrees of freedom
## Multiple R-squared:  0.9126, Adjusted R-squared:  0.9121
## F-statistic: 1858 on 1 and 178 DF,  p-value: < 2.2e-16
```

```
F_b <- summary(fs_b)$fstatistic["value"]
cat("First-stage F using RANK =", F_b, "\n")
```

```
## First-stage F using RANK = 1857.587
```

```
cat("First-stage R2 using RANK =", summary(fs_b)$r.squared, "\n")
```

```
## First-stage R2 using RANK = 0.9125559
```

```
# (c) Hausman test using first-stage residual from (b)
dat$vhat_rank <- resid(fs_b)

hausman_c <- lm(MSFT_RF ~ MKT_RF + vhat_rank, data = dat)
summary(hausman_c)
```

```
##
## Call:
## lm(formula = MSFT_RF ~ MKT_RF + vhat_rank, data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.27140 -0.04213 -0.00911  0.03423  0.34887
##
```

```
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.003018   0.005984   0.504   0.6146
## MKT_RF       1.278318   0.126749  10.085 <2e-16 ***
## vhat_rank    -0.874599   0.428626  -2.040   0.0428 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.08012 on 177 degrees of freedom
## Multiple R-squared:  0.3672, Adjusted R-squared:  0.36
## F-statistic: 51.34 on 2 and 177 DF,  p-value: < 2.2e-16
```

```
p_c <- summary(hausman_c)$coefficients["vhat_rank", "Pr(>|t|)"]
cat("Hausman p-value using RANK =", p_c, "\n")
```

```
## Hausman p-value using RANK = 0.04278906
```

```
# (d) IV using RANK
iv_d <- ivreg(
  MSFT_RF ~ MKT_RF |
  RANK,
  data = dat
)

summary(iv_d)
```

```
##
## Call:
## ivreg(formula = MSFT_RF ~ MKT_RF | RANK, data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.271625 -0.049675 -0.009693  0.037683  0.355579
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.003018   0.006044   0.499   0.618
## MKT_RF       1.278318   0.128011   9.986 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.08092 on 178 degrees of freedom
## Multiple R-Squared:  0.3508, Adjusted R-squared: 0.3472
## Wald test: 99.72 on 1 and 178 DF,  p-value: < 2.2e-16
```

```
confint(iv_d)
```

```
##              2.5 %      97.5 %
## (Intercept) -0.008827134 0.01486322
## MKT_RF       1.027421458 1.52921503
```

```
cat("OLS beta =", coef(ols_a)["MKT_RF"], "\n")
```

```
## OLS beta = 1.20184
```

```
cat("IV beta using RANK =", coef(iv_d)["MKT_RF"], "\n")
```

```
## IV beta using RANK = 1.278318
```

```
# (e) POS and first stage using RANK and POS  
dat$POS <- ifelse(dat$MKT_RF > 0, 1, 0)
```

```
fs_e <- lm(MKT_RF ~ RANK + POS, data = dat)  
summary(fs_e)
```

```
##  
## Call:  
## lm(formula = MKT_RF ~ RANK + POS, data = dat)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -0.109182 -0.006732  0.002858  0.008936  0.026652   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept) -0.0804216  0.0022622  -35.55  <2e-16 ***  
## RANK         0.0009819  0.0000400   24.55  <2e-16 ***  
## POS         -0.0092762  0.0042156   -2.20   0.0291 *    
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 0.01451 on 177 degrees of freedom  
## Multiple R-squared:  0.9149, Adjusted R-squared:  0.9139   
## F-statistic: 951.3 on 2 and 177 DF,  p-value: < 2.2e-16
```

```
test_e <- linearHypothesis(  
  fs_e,  
  c("RANK = 0", "POS = 0")  
)  
  
test_e
```

```
##  
## Linear hypothesis test:  
## RANK = 0  
## POS = 0  
##  
## Model 1: restricted model  
## Model 2: MKT_RF ~ RANK + POS  
##  
##    Res.Df    RSS Df Sum of Sq    F    Pr(>F)      
## 1      179 0.43784
```

```
## 2      177 0.03727  2    0.40057 951.26 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
cat("First-stage R2 using RANK and POS =", summary(fs_e)$r.squared, "\n")
```

```
## First-stage R2 using RANK and POS = 0.9148844
```

```
# (f) Hausman test using first-stage residual from (e)
dat$vhat_rank_pos <- resid(fs_e)

hausman_f <- lm(MSFT_RF ~ MKT_RF + vhat_rank_pos, data = dat)
summary(hausman_f)
```

```
##
## Call:
## lm(formula = MSFT_RF ~ MKT_RF + vhat_rank_pos, data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.27132 -0.04261 -0.00812  0.03343  0.34867
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.003004   0.005972   0.503   0.6157
## MKT_RF        1.283118   0.126344  10.156 <2e-16 ***
## vhat_rank_pos -0.954918   0.433062  -2.205   0.0287 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.07996 on 177 degrees of freedom
## Multiple R-squared:  0.3696, Adjusted R-squared:  0.3625
## F-statistic: 51.88 on 2 and 177 DF,  p-value: < 2.2e-16
```

```
p_f <- summary(hausman_f)$coefficients["vhat_rank_pos", "Pr(>|t|)"]
cat("Hausman p-value using RANK and POS =", p_f, "\n")
```

```
## Hausman p-value using RANK and POS = 0.02874032
```

```
# (g) IV using RANK and POS
iv_g <- ivreg(
  MSFT_RF ~ MKT_RF |
  RANK + POS,
  data = dat
)

summary(iv_g)
```

```
##
## Call:
```

```
## ivreg(formula = MSFT_RF ~ MKT_RF | RANK + POS, data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.27168 -0.04960 -0.00983  0.03762  0.35543
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.003004   0.006044   0.497    0.62
## MKT_RF       1.283118   0.127866  10.035 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.08093 on 178 degrees of freedom
## Multiple R-Squared:  0.3507, Adjusted R-squared:  0.347
## Wald test: 100.7 on 1 and 178 DF, p-value: < 2.2e-16
```

```
confint(iv_g)
```

```
##              2.5 %      97.5 %
## (Intercept) -0.00884329 0.01485031
## MKT_RF       1.03250520 1.53373109
```

```
cat("OLS beta =", coef(ols_a)["MKT_RF"], "\n")
```

```
## OLS beta = 1.20184
```

```
cat("IV beta using RANK =", coef(iv_d)["MKT_RF"], "\n")
```

```
## IV beta using RANK = 1.278318
```

```
cat("IV beta using RANK and POS =", coef(iv_g)["MKT_RF"], "\n")
```

```
## IV beta using RANK and POS = 1.283118
```

```
# (h) Manual Sargan test
dat$uhat_iv_g <- resid(iv_g)

sargan_reg <- lm(uhat_iv_g ~ RANK + POS, data = dat)
summary(sargan_reg)
```

```
##
## Call:
## lm(formula = uhat_iv_g ~ RANK + POS, data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.26914 -0.04702 -0.00801  0.03771  0.35674
```



```
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.0022220  0.0126326  -0.176    0.861
## RANK        0.0001370  0.0002234   0.613    0.540
## POS         -0.0174499  0.0235409  -0.741    0.460
##
## Residual standard error: 0.08103 on 177 degrees of freedom
## Multiple R-squared:  0.003103,    Adjusted R-squared:  -0.008162
## F-statistic: 0.2754 on 2 and 177 DF,  p-value: 0.7596
```

```
n <- nrow(dat)
R2_sargan <- summary(sargan_reg)$r.squared
Sargan_stat <- n * R2_sargan

df_sargan <- 2 - 1
p_sargan <- 1 - pchisq(Sargan_stat, df = df_sargan)

cat("Sargan statistic =", Sargan_stat, "\n")
```

```
## Sargan statistic = 0.5584634
```

```
cat("Sargan df =", df_sargan, "\n")
```

```
## Sargan df = 1
```

```
cat("Sargan p-value =", p_sargan, "\n")
```

```
## Sargan p-value = 0.45488
```

```
if (p_sargan < 0.05) {
  cat("Reject surplus IV validity at 5% level.\n")
} else {
  cat("Do not reject surplus IV validity at 5% level.\n")
}
```

```
## Do not reject surplus IV validity at 5% level.
```